Math 110 Hard Mock Midterm 2

50 minutes 4 questions 40 points

Set a 50-minute timer. Give concise reasoning with each answer. You may quote results from the notes or text if you state them. $\mathcal{P}_n(\mathbb{R})$ is the space of real polynomials of degree at most n. All bases are ordered.

1. Two subspaces of matrices (8 points) In $V = \mathbb{R}^{2,2}$ let

$$U = \left\{A : A\begin{pmatrix}1\\1\end{pmatrix} = 0\right\}, \qquad W = \{A : A^T = A\}.$$

a) (3) Find $\dim U$, $\dim W$, and a basis of $U \cap W$.

b) (3) Is $V = U + W$? Is the sum direct? Prove both answers.

c) (2) Find a subspace $W' \subseteq W$ with $V = U \oplus W'$, and prove it works.

2. Factoring a map (10 points) Let V, W, X be finite-dimensional, and let $S\colon V \to X$ and $T\colon V \to W$ be linear.

a) (2) Prove: if $T = RS$ for some $R \in \mathcal{L}(X, W)$, then $\operatorname{null} S \subseteq \operatorname{null} T$.

b) (3) Now assume $\operatorname{null} S \subseteq \operatorname{null} T$. For $y \in \operatorname{range} S$, define $R_0(y) = Tv$, where v is any vector with $Sv = y$. Prove that $R_0\colon \operatorname{range} S \to W$ is well defined and linear.

c) (3) Prove that there is $R \in \mathcal{L}(X, W)$ with $T = RS$.

d) (2) Let $S\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_1(\mathbb{R})$ be $S(p) = p''$. For $T(p) = p'(1) - p'(0)$, find an explicit R with $T = RS$. For $T(p) = p(0)$, prove no such R exists.

3. Nonstandard bases on both sides (10 points) Define $T\colon \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$ by

$$T(p) = \big(p(1),\ p'(1),\ p(2)\big).$$

Use the basis $B = \big(1,\ x-1,\ (x-1)^2\big)$ of the domain and $C = \big((1,0,0),(1,1,0),(1,1,1)\big)$ of the codomain.

a) (4) Find $\mathcal{M}(T)$ with respect to B and C.

b) (3) Prove that T is an isomorphism without using the matrix.

c) (3) Find the $p \in \mathcal{P}_2(\mathbb{R})$ with $T(p) = (1,1,1)$, and write it in the standard basis.

4. Preimages through a quotient (12 points) Let V and W be finite-dimensional, $T \in \mathcal{L}(V, W)$, and U a subspace of W. Define

$$S\colon V \longrightarrow W/U, \qquad S(v) = Tv + U.$$

a) (3) Prove that S is linear and that $\operatorname{null} S = T^{-1}(U)$.

b) (3) Prove that $\dim \operatorname{range} S = \dim(\operatorname{range} T + U) - \dim U$.

c) (3) Deduce $\dim T^{-1}(U) = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T)$.

d) (3) Let $D\colon \mathcal{P}_4(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ be differentiation and $U = \operatorname{span}(x^3, x^4)$. Use (c) to find $\dim D^{-1}(U)$, then give a basis of $D^{-1}(U)$.

Solutions

Question 1

(a) $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is in U exactly when $a+b=0$ and $c+d=0$. U is the null space of $A \mapsto A(1,1)^T$, which is surjective onto $\mathbb{R}^2$ ($E_{11} \mapsto e_1$, $E_{21} \mapsto e_2$), so $\dim U = 4-2=2$. $\dim W = 3$ (basis $E_{11}, E_{22}, E_{12}+E_{21}$). $U \cap W$: symmetric with $a+b=0$ and $b+d=0$, so $A = \left(\begin{smallmatrix} a & -a \\ -a & a \end{smallmatrix}\right)$. Basis: $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

(b) $\dim(U+W) = 2+3-1 = 4 = \dim V$, so $U+W=V$. Not direct, since $U \cap W \neq \{0\}$. (Trap: a sum can be all of V without being direct. Here $2+3>4$ forces a nonzero intersection before you compute anything.)

(c) $W' = \operatorname{span}(E_{11}, E_{22})$, the diagonal matrices. A diagonal matrix $\operatorname{diag}(a,d)$ is in U only if $a=0$ and $d=0$, so $U \cap W' = \{0\}$; and $2+2=4$. So $V = U \oplus W'$.

Question 2

(a) If $Sv=0$ then $Tv = R(Sv) = R(0) = 0$.

(b) Well defined: if $Sv = Sv' = y$, then $v - v' \in \operatorname{null} S \subseteq \operatorname{null} T$, so $Tv = Tv'$. Linear: if $Sv = y$ and $Sv' = y'$, then $S(\lambda v + v') = \lambda y + y'$, so $R_0(\lambda y + y') = T(\lambda v + v') = \lambda R_0(y) + R_0(y')$.

(c) Take a basis $y_1, \ldots, y_r$ of $\operatorname{range} S$ and extend it by $x_1, \ldots, x_s$ to a basis of X. Define $R(y_i) = R_0(y_i)$ and $R(x_j) = 0$ (LADR 3.4). Then R agrees with R_0 on $\operatorname{range} S$ (both linear, agree on a basis), so $RSv = R_0(Sv) = Tv$ for every v. (Trap: R_0 alone is not an answer; it is only defined on $\operatorname{range} S$, not on X.)

(d) $R(q) = \int_0^1 q(t)\,dt$ works: $R(p'') = \int_0^1 p'' = p'(1) - p'(0)$. For $T(p) = p(0)$: $1 \in \operatorname{null} S$ but $T(1) = 1 \neq 0$, so $\operatorname{null} S \not\subseteq \operatorname{null} T$, and by (a) no R exists.

Question 3

(a) Images: $T(1) = (1,0,1)$, $T(x-1) = (0,1,1)$, $T\big((x-1)^2\big) = (0,0,1)$. In C: $(\alpha,\beta,\gamma) = a(1,0,0) + b(1,1,0) + c(1,1,1)$ gives $c=\gamma$, $b=\beta-\gamma$, $a=\alpha-\beta$. So

$$\mathcal{M}(T) = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}.$$

(b) Both spaces have dimension 3, so it is enough to show injectivity. If $T(p)=0$, then $p(1) = p'(1) = 0$, so $(x-1)^2$ divides p: $p = c(x-1)^2$. Then $p(2) = c = 0$. So $p=0$.

(c) Write $p = a + b(x-1) + c(x-1)^2$. Then $p(1) = a = 1$, $p'(1) = b = 1$, $p(2) = a+b+c = 1$, so $c=-1$. $p = 1 + (x-1) - (x-1)^2 = -x^2 + 3x - 1$. Check: $p(1)=1$, $p'(x) = -2x+3$ so $p'(1)=1$, $p(2) = -4+6-1 = 1$.

Question 4

(a) S is T followed by the quotient map $\pi\colon W \to W/U$, a composition of linear maps. $S(v) = 0 + U$ exactly when $Tv \in U$, i.e. $v \in T^{-1}(U)$.

(b) $\operatorname{range} S = \{Tv + U\} = \pi(\operatorname{range} T)$. Since $\pi(u) = 0$ for $u \in U$, also $\pi(\operatorname{range} T + U) = \pi(\operatorname{range} T)$. So $\operatorname{range} S$ is the range of π restricted to $X = \operatorname{range} T + U$. The null space of that restriction is $X \cap U = U$. Rank–nullity on X: $\dim \operatorname{range} S = \dim(\operatorname{range} T + U) - \dim U$.

(c) Rank–nullity for S:

$$\dim T^{-1}(U) = \dim V - \dim(\operatorname{range} T + U) + \dim U.$$

Substitute $\dim(\operatorname{range} T + U) = \dim \operatorname{range} T + \dim U - \dim(U \cap \operatorname{range} T)$ and $\dim V - \dim \operatorname{range} T = \dim \operatorname{null} T$.

(d) $\operatorname{null} D = \operatorname{span}(1)$. $\operatorname{range} D = \mathcal{P}_3(\mathbb{R})$ (the codomain is $\mathcal{P}_4$, and x^4 is not a derivative of anything in $\mathcal{P}_4$). $U \cap \operatorname{range} D = \operatorname{span}(x^3)$. So $\dim D^{-1}(U) = 1 + 1 = 2$. Basis: $p' \in \operatorname{span}(x^3, x^4)$ with $\deg p' \leq 3$ means $p' = cx^3$, so $p = a + \frac{c}{4}x^4$. Basis $1,\ x^4$. (Trap: $U \cap \operatorname{range} D$ is not U here, unlike the practice midterm, because the codomain is $\mathcal{P}_4$.)